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User behaviors analysis on OTT platform with an integration of technology acceptance model

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Abstract

This research is based on the theory of Technology Acceptance Model to understand consumer behaviors on paid video streaming platforms and explore whether perceived price and user age will moderate user behaviors. Another focus topic is the pricing mechanism of paid video streaming platforms in Taiwan, and we find that most users prefer recurring charges and hybrid plan, which is consistent with the results of the most commonly used paid video streaming platform. The research results prove that when consumers are more satisfied with their personal motivation and system quality, they will increase their satisfaction with perceived ease of use alongside perceived usefulness, thereby improving their overall attitude towards the use of the platform and increasing their willingness to use. In this research, users' age does not contribute to the difference between attitude toward use and actual use behavior. If one holds a positive attitude towards paid video streaming platforms, regardless of age, it will have a positive intention with continued use behavior in the future. Although users' age does not significantly moderate to the relationship between attitudes toward user behaviors, it can still be applied as a distinction for judging the intention to use. Finally, based on the research results, practical suggestions are provided to practitioners of OTT platforms.

Keywords Technology acceptance model · OTT platform · Personal motivation · System quality · Perceived construct · Behavioral construct

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1 Introduction

With the development of the television industry and technology, our viewing habits have undergone a sea-level change. Since Over-The-Top (OTT) services were introduced in the 2000s, consumer viewing habits and choices have changed. It is known as a streaming TV service includes any kind of video communication platform with access to high-speed internet connections, including cable, phone and satellite companies. OTT revolutionizes how content is distributed and how content is created. It involves numerous Internet distributions, as well as the spending for at-home entertainment (Snyman and Gilliard 2019). With its advanced technology, contents and convenience, more and more media giants such as Disney+, Apple tv+, Amazon Prime Video, Netflix, YouTube Premium, etc. are facing fierce competition and continuously rolling out more services and contents to appeal consumers. OTT also has drove traditional television companies to archiving more films and adopting new technology. However, eMarketer (2020) estimates that the number of “Cord-Cutters” in the United States has reached 31.2 million in 2020 and will continuously rise in the future: by 2024, more than one-third of the U.S. families will have opted out of their cable television subscriptions. According to a research report that is released by the Motion Picture Association of America, in 2018, the subscribers of OTT around the globe have officially been more than those of cable television (MAPP, 2018).

The pricing mechanism of paid video streaming platforms plays a crucial role in running business, which will be broken down to seven parts: recurring charges (monthly rent and all-inclusive package), hybrid plan (combination of recurring charges and payment access to single), single rental (rent by single movie), permanent access to a single movie, period access to a series, subscription to a specific genre, and cross-platform bundled package. If viewers consider the perceived price of any paid video streaming platforms to be cheap and cost-effective, their willingness to consume may be more triggered. In addition, based on the survey results of the 2020 Media Book, the viewing ratio of Taiwanese television viewers are proportional to their age, in which more than half of the viewers are shown to be more than forty-five years old (Taipei Association of Advertising Agencies 2020). Taiwan Network Information Center (2019) notes that Taiwanese internet users have outnumbered 20 million for the first time. The number of the users who watch audiovisual content on the internet is as high as 86.5%, and the usage rate is inversely proportional to their age. The usage rate of 15-to-29 year’s old viewers is over 90%. However, when the age of the users increases, the usage rate will slightly decrease. Only 50% of the 50-year-old-and-above viewers say that they will use the internet to watch programs and dramas, 20% keep the habit of watching movies, and about 10% never watch video-related content online. As for the payment status of various entertainment activities, the overall payment rate is 24%, the payment ratios for watching programs drama and movie programs are 10.01% and 11.01% respectively, and those of the 20-to-39 age group are the highest among all. Regarding the aforementioned statistics, the middle-aged and senior groups are more likely to watch cable television, while younger viewers tend to watch programs on the internet.

In the era of Internet of Things and big data, the innovative technology is expected to open a new area of business paradigm with huge opportunities (Chatterjee 2020). More studies are developed to cover attitude and behavioral intention of the consumers to adopt new technology (Chatterjee 2021). OTT streaming services are undoubtedly one of those innovative services. As far as practical observation is concerned, consumers’ viewing habits and audio-visual culture have changed accordingly. However, in addition to continuously introducing more services and contents to attract consumers, whether the quality and quantity of software and hardware services provided by OTT can affect their attitudes, behavioral images and actual use behaviors through consumers' beliefs. In other words, how to truly improve consumers' acceptance of OTT and provide important consumer behavior information in the course of OTT development is also a topic of great concern to OTT operators. This study mainly focuses on viewers' interaction with paid video streaming platforms and attempts to explore whether subjective perception of the price will affect their user behaviors. We adopt the Technology Acceptance Model (TAM) as the theoretical framework to dissect viewers' motivation of using paid video streaming platforms. The motivation is divided into to two types: personal motivation and system quality. By means of probing into viewers' motivation of using paid video streaming platforms, this study hopes to attain a deeper understanding of how they, when choosing a platform, think and behave. Additionally, this study is also helpful as a future guide to the paid video streaming platform industry. The first objective of this study is to understand the motivation that influences how viewers use paid video streaming platforms. Next, we study the perceived price affects to user attitudes and Actual use behaviors; and then the final objective is to understand whether age is also associated with the relationship between user attitudes and Actual use behaviors.

2 Literature review and research hypotheses

OTT is generally defined as the content, service, or application which is directly delivered by one software to viewers through the internet, where competitive low prices, enhanced customer experience, innovative service plans, content localization, strategic partnerships, flexibility in technology adoption, and aggressive sales promotion can drive consumer adoption of OTT from occasional to habitual (BEREC 2016; Sharma and Lulandala 2022). Therefore, the motivation that influences viewers using paid video streaming platforms can be fitted as content, price, flexibility, convenience (perceived ease of use), perceived usefulness, perceived enjoyment (hedonic motivation), desire to be free from any constraints, entertainment value, socialization, cultural inclusion, carnival, and self-induced adoption the efficacy of OTT streaming platforms (Mulla 2022). OTT platforms provide many innovative services, information content and technologies, but we need to deeply understand whether these are really what the audience cares about, accepts, and furthermore affects the changes in their using behavior. In light of TAM is broadly applied to attain in-depth understanding of user acceptance in information management research, the causal relationship between viewers' motivation and influences in this study is established based on this model. TAM, as shown in Fig. 1, is mainly applied to observe the causal relationship between external variables through belief variables of Perceived Usefulness and Perceived Ease of Use on Attitude toward Using, Behavioral Intention of Use and Actual use behavior. User Attitude is defined by the combination of Perceived Usefulness and Perceived Ease of Use (Taylor and Todd 1995). Influence users' attitudes towards the actual use of technology by enhancing users' behavioral intentions have been applied in the education (Chatterjee et al. 2021a). Therefore, TAM is very suitable to explain the usage behavior of OTT platform (Chatterjee et al. 2021b).

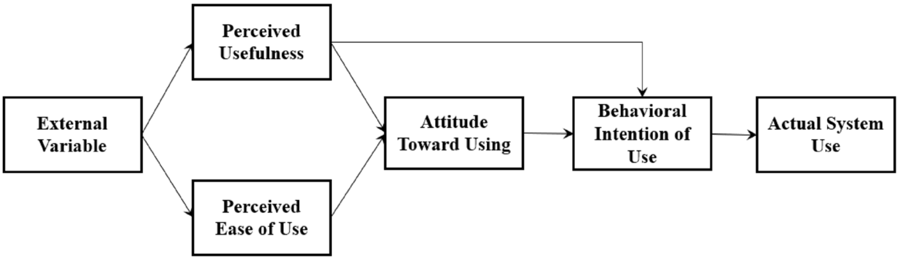

Fig. 1 Technology acceptance model structure

2.1 OTT customer motivation

In the framework of TAM, external variables may affect perceived usefulness and perceived ease of use. The external variables of this study include the viewer's psychological perception of personal motivation. Perceived Usefulness is the degree to which users believe that, when a specific system is adopted, their job performance can be improved. Perceived Ease of Use refers to the degree to which users believe that when a particular system is adopted, it can save their time, cost and workload. Therefore, the easier the system is to use, the more comfortable the users will feel to use the system. In terms of personal motivation, based on past research on viewer's needs, motivations and attitudes towards media, this study proposes entertainment, escape from reality, social contact, pastime, idolatry, information, and brand as variables that affect personal motivation. Chung et al. (2014) point out that information and personal intrinsic motivation are important to media selection. Entertainment can meet viewers' needs for pleasure, satisfaction, and relaxation (Wolfinbarger and Gilly 2001; Kim and Johnson 2012; Chong 2013). Leung and Wei (2000) find that motivation can be seen as an internal driving force to consumers' choice of media tools and greatly affect the intensity and persistence of user behaviors. Escapism enables people to temporarily run away from the overwhelming reality and is widely applied to further behavioral studies on internet and online gaming addiction (Lombard and Ditton 1997). Escapists expect to retrieve temporary relaxation and stress relief through such a distraction (Park et al. 2011; Swanson et al. 2008). Diddi and Larose (2006) verify that escapism is directly proportional to internet use. Park et al. (2011) confirm that internet is positively associated with escapism and readers, especially when waiting, can attain escapism through paid video streaming platforms as time killers to avoid boredom. The influence of social interaction is a subjective awareness of whether the most significant others think an individual should or shouldn't partake certain behaviors (Davis et al. 1989). Yoo et al. (2020) studies the consumption motivation behind audiovisual content and verifies that nearsightedness is associated with enjoyment, efficiency, social interaction, and information seeking. When viewers suffer from nothing to do, they can take good advantage of social networking sites to handle their free time (Pornsakulvanich and Dumrongsiri 2013). Papacharissi and Rubin (2000) point out those viewers are more intimate to the companies of online media tools which entitle them with time killing through paid video streaming platforms. The idols are open to be heroes, celebrities, or role models (Maltby et al. 2004), pop stars or athletes (He 2006). Idolatry, especially of pop stars and movie stars therefore serves as an unneglectable research area due to its prevalence, association with peer norms, and impact on youth development (Maltby et al. 2004; McCutcheon et al. 2003). Information is a crucial indicator of a website's competitiveness that affects user behaviors (Sujata et al. 2015). Holzwarth et al. (2006) state that what information a website reveals serves as a crucial motivation for users. If users are satisfied with the amount of the information, they are more likely to increase the willingness to continue adopting the information (Redmond 2002). Adhering to the principle of life-long learning, constantly updating the amount of self-information is ensuring to an individual's competitiveness, Leung (2003) emphasizes the significance of taking information motivation into consideration in his research. Brand image is a subjective attitude of consumers toward a brand. If a brand clearly fathoms its target groups and their purposes of using an app, OTT platform marketing will be facilitated to operate more effectively to communicate with the target groups (Kim and Shin 2017; Dasgupta and Grover 2019; Amoroso et al. 2021). Regarding the aforementioned criteria, we believe that personal motivation is positively associated with perceived usefulness and perceived ease of use. The hypothesis is proposed as follows:

H1: Personal motivation influences Perceived usefulness.
H2: Personal motivation influences Perceived ease of use.

2.2 OTT system quality

One of the differences between paid video streaming platforms and general free video streaming platforms is the video quality. This study proposes stability, function, interface design, convenience, real-time, diversity, originality, security, norm as variables comprised system quality. Stability is an important factor to reflect the streaming technology of OTT platform. If the videos are flawed by download delay, it may affect the confidence in systemic stability (Ahn et al. 2004). The practical functions, including search function, recommendation system, viewing history, resumption of playback from breakpoints, sharing by multiple people, simultaneous playback on multiple screens, and offline viewing, could affect he willingness to pay for OTT service (Kim et al. 2017; Kwon et al. 2020). Interface design refers to the process, in which designers construct software for users to deem it to be practical and user-friendly. Sujata et al. (2015) pinpoints that factors, such as cost, convenience, features, social propensity, content availability, smartphone and mobile internet penetration, user experience, and neutrality which are effectively leveraged by OTT players can give birth to better service. Convenience refers that OTT service are no longer limited by devices and time. Almost all respondents agree that the most crucial reasons of this change are the convenience of service, availability of personal media and international content (Wolk 2015; Singh 2019). Real-time refers to the degree to which a system responds immediately to the behavior of a request for information or data (Wixom and Todd 2005). Shin et al. (2016) find that consumers apparently prefer live broadcast service, terrestrial TV content, latest broadcasts/movies, and an increase in the number of available VODs. Park (2018) states that live TV shows are extremely popular in many countries.

Media diversity aims to provide diverse content, viewpoints and content types, so that consumers, when watching movies and getting access to the opinions from different social classes, will be blessed with richer choices. While some platforms try to cater to the majority of audiences, and choose to ignore the preferences of relatively minority groups, resulting in the gradual similarity of traditional media products (Cha 2014; Van der Wurff 2003), others find that when viewers are using music service in mobile phones, content diversity and quality, rather than price, are more valued (McQuail 1992; Vlachos 2009; Cha 2014). OTT platforms create and deliver Originality which are more engaging content unavailable elsewhere (Dasgupta and Grover 2019). Most of Netflix advertised content is so original that should be more accurately described as exclusive in a particular market.

Netflix spends a crucial portion of its content budget on original programming (Lotz 2016; Wayne 2018). Security is crucial when users conduct transactions in virtual markets. They may be doubtful of personal information security due to the uncertainty of network security (Liebermann and Stashevsky 2002). Samuelson (1990) addresses that digital media are characterized by convenient reproduction, independence of digital work transmission, simultaneous use by multiple people, mass storage in digital mode, etc. Therefore, video streaming platforms involve issues of privacy. Norm as the degree to which viewers believe that the platform needs to be regulated by laws and the degree to which they can abide by relevant norms when they use paid video streaming platforms. Bilbil (2018) believes that it is recommended for OTT platforms to ensure privacy, data protection, price control, effective competition and appropriate tax regulations. Regarding the above discussions, system quality is believed positively associated with perceived usefulness and perceived ease of use. The hypothesis is proposed as follows:

H3: System quality influences Perceived usefulness.
H4: System quality influences Perceived ease of use.

In addition, TAM also can be applied to observe the causal relationship between Perceived Usefulness and Perceived Ease of Use on Attitude Toward Using, Behavioral Intention of Use and Actual use behavior. User Attitude is defined by the combination of Perceived Usefulness and Perceived Ease of Use (Taylor and Todd 1995). Influence users' attitudes towards the actual use of technology by enhancing users' behavioral intentions have been applied in the education (Chatterjee Bhattacharjee et al. 2021a, b). This study is based on the theory of TAM to understand consumer behaviors on paid video streaming platforms. Therefore, the TAM is applied to explain the usage behavior of viewers using paid video streaming platforms. When the external variables referred to OTT customer motivation and system quality in this study, the hypotheses H5 to H9 are therefore established according to TAM. The hypotheses are proposed as follows:

H5: Perceived usefulness influences Actual use behavior.
H6: Perceived usefulness influences Attitude.
H7: Perceived ease of use influences Perceived usefulness.
H8: Perceived ease of use influences Attitude.
H9: Attitude influences Actual use behavior.

2.3 Perceived price

Perceived price is positively affected by pricing goals alongside quality perception and negatively affected by the reference price which is a fabricated one by consumers through subjective judgements (Chang and Wildt 1994; Kashyap and Bojanic 2000). Consumers do not always remember the actual price they pay, and what is really meaningful to consumers is the perceived price defined in their minds (Zeithaml 1988). Lichtenstein et al. (1993) believe that the definition of perceived price also includes consumers' subjective judgements and preference. Age sometimes influences consumer behaviors. In research focusing India, the top OTT is becoming a preferred source of entertainment amongst young consumers over traditional Pay TV service (Cable TV/DTH) (Sadana and Sharma 2021). While the determination and ranking of factors that are important in considering a subscription of an OTT platform service could differ from generation to generation (Koul et al. 2020), the perceived price and age are assumed to be the interference variables between attitude and actual usage behavior. The perceived price and age are assumed to be interference variables between attitude and actual usage behavior. The hypothesis is proposed as follows:

H10: Perceived price interferes with Attitude and Actual use behavior.
H11: Age interferes with Attitude and Actual use behavior.

3 Methodology

3.1 Research model

This study constructs a theoretical framework shown in Fig. 2, which can explain the behavioral motivation of viewers who use paid video streaming platforms. Personal motivation and System quality, as external variables in the TAM model, interact with perceptual aspects such as Perceived usefulness and Perceived ease of use, which in turn affect user Attitude and Actual use behavior, and add Perceived price and Age as moderating variables (Fig. 3).

3.2 Data collection

The questionnaires are distributed on Facebook questionnaire exchange clubs and social platforms which are commonly used by college students. Friends and relatives are invited to share the questionnaire link. The questions are designed with reference to the research of domestic and foreign scholars in related fields. The subjects answer anonymously. The total number of questions is 82. The aspects of the questionnaire are named as follows: User experience (which platforms have been used, which platforms have been used most often, contact time, devices used, frequency of use, time of each use, and sharing conditions), Usage factors (usage motivation, usage behavior, idolatry, brand image and safety, etc.), Perception and Behavior (Explore users' attitudes and behaviors toward paid video streaming platforms, as well as their perceptions of the ease of use and usefulness of the platforms. A score range of 1–5 was given according to the level of agreement, representing "strongly disagree", "disagree", "normal", "agree" and "strongly agree"), Perceived price and pricing mechanism (Perceived price is to explore whether users' sensitivity to price

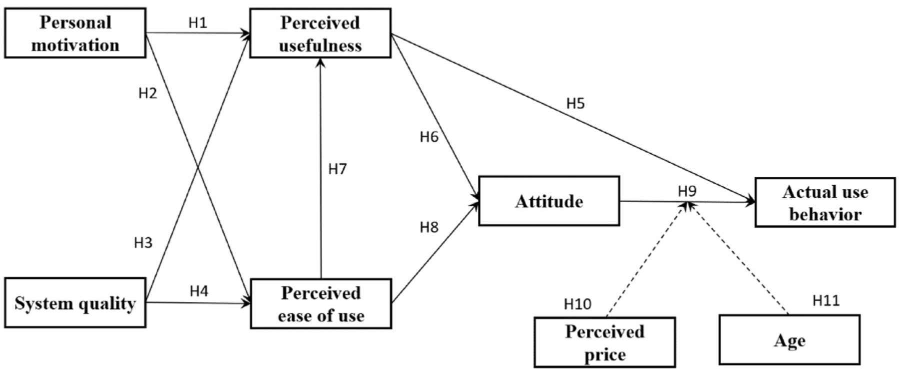

Fig. 2 The theoretical framework

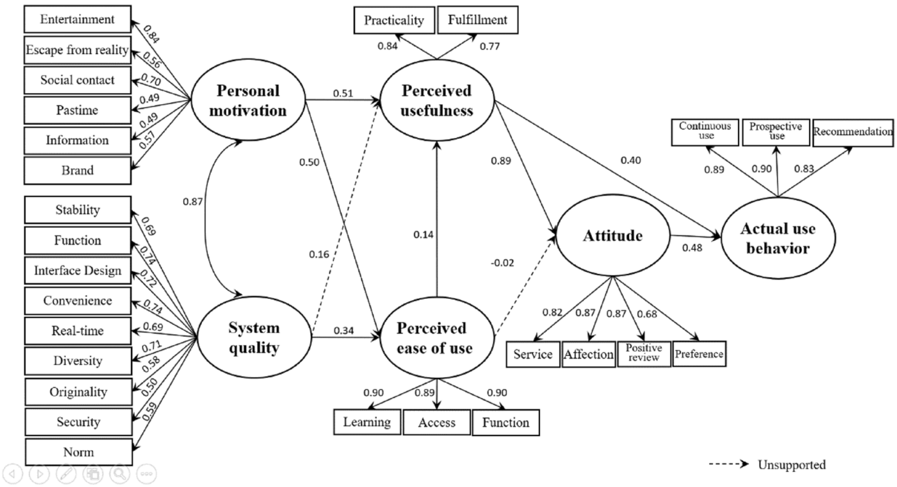

Fig. 3 Model structure

will affect their subscription willingness. Pricing mechanism is to explore readers' preferences for existing mainstream pricing mechanisms. Both are measured by using the 5-Point Likert Scale), and Demographic variables (gender, age, marital status, occupation, average monthly income, education level and area of residence).

As a notion of comprised questions, the subconstruct stands for the perspective, experience or consequences about using OTT. In Personal motivation, the respondent is: being delighted, entertained or having fun (Entertainment); temporarily forgetting the worries, decompressed, getting away from responsibility and work (Escape from reality); in need of getting through waiting time or leisure time (Pastime); getting closer and more understanding of idols (Idolatry); having more talking points, fostering interaction and relationships among peers (Social contact); better understanding current events and daily affairs, accruing more information (Information); aware of the importance of the brand and its impact on subscription (Brand). In System quality, the platform provides: high-definition films, acoustic fidelity, less glitches, translation accuracy, quick response (Stability); easy-to-search function, recommendations, viewing history, cross-device, simultaneous and offline watching (Function); humanized interface, genre design, simplified interface (Interface); flexibility of time, places and devices for watching (Convenience); newest content and information, more latest films, swift updating, prompt dramatic series releasing (Real-time); versatile and a great volume of content (Diversity); original or exclusive content (Originality); information and payment security, protection on audience rights (Security). The Norm questions are user's recognition on disclosure revealing, regulation obedience, necessity of the platform being regulated.

3.3 Reliability

This study conducts a preliminary statistical analysis of demographic variables to understand the sample distribution and then performs a descriptive statistical analysis for each construct for the status of each section. Reliability analysis can be applied to examine the internal homogeneity and stability of the scale. This study adopts the Cronbach's alpha coefficient measurement method in the Coefficient of Internal Consistency, which is also the most commonly used reliability in the Likert scale. Confirmatory factor analysis (CFA) examines the fitness of the model through the comparison of factor loading to define the influence and relationship between the variables. In this study, CFA was applied to examine the convergent and discriminant validity of the scale. Structural Equation Modeling (SEM) is a causal relationship analysis model, which measures the relationship between dependent variables and latent variables, evaluates direct and indirect effects, and can also be applied to detect the interference effects of interference variables. In this study, it explores the causal relationship between personal factors, system quality alongside actual use behavior, and the interference effect of age and perceived price.

4 Analysis and results

An online questionnaire is conducted from October to November in 2020. The questionnaires are distributed through the communication software LINE and the Facebook questionnaire exchange community. 832 questionnaires were gathered in total, and 821 valid samples were obtained.

4.1 Descriptive statistics

Most of the respondents are female (75.76%), 19-to-25 years old (40.32%), unmarried (66.75%), students (33.13%), monthly income below NT$5,000 (19.37%), and university or college degree (71.99%). Perceived price is an interaction variable which is set in this study, in which whether "the users subscribe to the streaming platform or not depends on the price" has the highest recognition score. In light of the pricing mechanism, the "Recurring charges" plan accounts for 73.2% of the total and is followed by the "Hybrid plan" (15.3%), which is in resonance with the situation today.

4.2 Reliability analysis

The subconstructs are examined by the absolute value of skewness and kurtosis coefficient, and it can be reasonably inferred that they conform to the normal distribution. Among the subconstructs of Personal motivation, Pastime has the highest score (4.27). The most crucial subconstruct in System quality is Convenience, followed by Diversity, which means that users of paid video streaming platforms pay more attention to the convenience of their systems and the diversity of content. However, the scores for Security and Norm are relatively low, which indicates that users still have doubts about these two aspects. Compared with other constructs, the items of perception have a higher degree of recognition, and it can be inferred that users think that paid video streaming platforms are useful and user-friendly. The scores of the Attitude and Actual use behavior are not obviously different, it can be inferred that users will continuously use paid video streaming platforms in the future, and overall, they are satisfied with the service of the platforms. The reliability of Personal motivation and System quality are 0.905 and 0.921 respectively, both as excellent reliability. The reliability of the Perceived usefulness, Perceived ease of use, Attitude, Actual use behavior is also dwells on high reliability level; the total reliability of this questionnaire is 0.958 with higher reliability (Table 1).

4.3 Confirmatory factor analysis (CFA)

According to Bollen (1989), if the Mardia coefficient is less than p(p + 2) (*p* is the number of observed variables), the sample size has multivariate normality. The Mardia coefficient of the measurement model in this study is 202.848. This scale contains 27 observed variables, with a p(p + 2) of 783, which confirms that the measurement model conforms to multivariate normality. The fit chi-square value of the measurement model is 704.378, and the probability *P* is less than 0.05 which indicates that the conceptual model of this study is not well suited to the actual data. In this measurement model, the ratio of the chi-square value to the degrees of freedom is 2.28, which is less than the standard value of 3 (Bagozzi and Yi 1988). The solid measurement model and the data are well matched. Composite Reliability (CR value) all meets the standard of 0.6; Average of Variance Extracted (AVE) of each aspect is between 0.38 and 0.80. Although it does not meet the standard of 0.5, it is still an acceptable range. Overall, this measurement model generally meets the criteria of convergent validity. This study applies the confidence interval method, which is proposed by Torkzadeh, Koufteros et al. (2003), at the 95% confidence level (Table 2).

4.4 SEM analysis and results from hypothesis testing

When the confidence interval of the coefficient is established in the SEM, the Bootstrap estimation method is applied. If the confidence interval doesn't contain 1, it means that it is highly related, which indicates that the dimensions have discriminant validity. In this study, the Bootstrap estimation method was used to repeat the estimation 1000 times. After examining the violation estimation pattern (Hair et al. 2009), there is no violation estimation in this structural model. The Bollen-Stine p-value correction method proposed by Bollen and Stine (1992) is applied in this study. After 2000 Bootstraps, the overall model fit measures including absolute, parsimonious and relative indexes match the acceptable value standard. The structural model is well matched (Table 3).

According to the analysis results, the hypothesis testing of the research is carried out. Overall, 7 of the 9 hypotheses of the validation model reach a significant level, while Hypotheses 3 and 8 could not be established. The hypothesis H1 is established. It means that the stronger the personal motivation to use paid video streaming platforms is, the higher the usefulness which is felt by the user will be. The hypothesis H2 is established as well, which means that the stronger the personal motivation of using the platform is, the higher the ease of use the users will feel. The hypothesis H3 is not supported. That the users put great emphasis on the system quality does not affect perceived usefulness. The hypothesis H4 is established, which means that the higher the attention is paid to the system quality of paid video streaming platforms, the more clearly the ease of use the users can be felt. The supported result of hypothesis H5 means that the higher the usefulness of using paid video streaming platforms is thought by the users, the higher the frequency will be for the users to continuously use them in the future. The established result of hypothesis H6 shows that the higher the usefulness is, the more positive the user's attitude towards the platform will be. The sustained result of hypothesis H7 indicates that the higher the ease of use is felt when the users use paid video streaming platforms, the higher the usefulness the users will feel. The result of hypothesis H8 is unsupported, which implies that the higher the ease of use is felt when the users use the platform, the more positive the user's attitude towards the platform will be. The supportive result of hypothesis H9 explains that the

Table 1 Measurement model evaluation

Construct	Scores	Cronbach's Alpha	Construct	Scores	Cronbach's Alpha	Construct	Scores	Cronbach's Alpha
Personal motivation		0.905	System quality		0.921	Perceived usefulness	4.15	0.820
Entertainment	4.20	0.889	Stability	3.87	0.710	Perceived ease of use	4.25	0.924
Escape from reality	3.81	0.890	Function	3.97	0.673	Attitude	3.96	0.874
Pastime	4.27	0.856	Interface Design	3.90	0.855	Actual use behavior	4.03	0.904
Idolatry	3.12	0.912	Convenience	4.23	0.901			
Social contact	3.55	0.923	Real-time	3.89	0.785			
Information	3.45	0.881	Diversity	4.04	0.858			
Brand	4.00	0.744	Originality	3.96	0.835			
			Security	3.44	0.609			
			Norm	3.86	0.699			

Table 2 Validation analysis for measurement models

Construct	SFL	t-Value	SE	SMC	EV	CR	AVE
Personal motivation						0.78	0.38
Entertainment	0.833	21.236	0.11	0.69	0.12		
Escape from reality	0.557	12.326	0.16	0.31	0.43		
Pastime	0.699	16.512	0.13	0.49	0.22		
Social contact	0.489	10.591	0.16	0.24	0.46		
Information	0.496	10.779	0.16	0.25	0.44		
Brand	0.561	12.498	0.12	0.32	0.23		
System quality						0.88	0.44
Stability	0.692	16.441	0.12	0.48	0.23		
Function	0.736	17.873	0.10	0.54	0.13		
Interface design	0.722	17.371	0.11	0.52	0.18		
Convenience	0.739	17.967	0.13	0.55	0.22		
Real-time	0.694	16.453	0.13	0.48	0.23		
Diversity	0.706	16.852	0.13	0.50	0.23		
Originality	0.577	13.026	0.15	0.33	0.38		
Security	0.495	10.874	0.14	0.25	0.34		
Norm	0.588	13.33	0.12	0.35	0.22		
Perceived usefulness						0.82	0.70
Perceived ease of use						0.92	0.80
Attitude						0.88	0.66
Actual use behavior						0.91	0.77
Mardia coefficient	202.848			$p(p+2)$	$=783$	0.97	0.56

Table 3 Overall model fit index

Fit measures	Acceptable value	Model testing results
(After Bollen-Stine Bootstrap)		
Absolute		
χ^2 ($p=0$)	The smaller the better ($P \geq \alpha$)	391.388
χ^2/df	<3	1.246
GFI	>0.9	0.944
AGFI	>0.9	0.933
RMR	<0.08	0.021
SRMR	<0.08	0.041
RMSEA	<0.08	0.023

more positive the user team's attitude towards using paid video streaming platforms is, the higher the frequency will be for the users to continuously use them in the future (Table 4).

Table 4 Hypothesis test results

Hypothesized path	Coefficient	T-values	Remarks
H1 Personal motivation → Perceived usefulness	0.508***	4.282	Supported
H2 Personal motivation → Perceived ease of use	0.495***	4.521	Supported
H3 System quality → Perceived usefulness	0.155($P=0.13$)	1.913	Unsupported
H4 System quality → Perceived ease of use	0.336*	3.348	Supported
H5 Perceived usefulness → Actual use behavior	0.401**	4.115	Supported
H6 Perceived usefulness → Attitude	0.886***	6.193	Supported
H7 Perceived ease of use → Perceived usefulness	0.338*	4.597	Supported
H8 Perceived ease of use → Attitude	− 0.018($P=0.942$)	− 0.195	Unsupported
H9 Attitude → Actual use behavior	0.481***	5.157	Supported

*Represents $p<0.05$, **represents $p<0.01$, ***represents $p<0.001$

4.5 Effect analysis

Personal motivation has a positive direct (0.495) effect on Perceived ease of use, and has a positive direct (0.508) and indirect (0.168) effect on Perceived usefulness through Perceived ease of use, with a total effect of 0.675. The indirect effect of Personal motivation on Attitude is 0.589. The indirect effect of Personal motivation on Actual use behavior is 0.554. System quality has a direct (0.155, insignificant) effect on perceived usefulness, an indirect effect value through perceived ease of use is 0.114, and the overall effect value is 0.269. The direct effect of System quality on Perceived ease of use is 0.336. The indirect effect values on Attitude and Actual use behavior are 0.232 and 0.219 respectively. The direct effect of Perceived ease of use on Perceived usefulness is 0.338. Perceived ease of use has direct (− 0.018, not significant) and indirect (0.300) effects on Attitude, with a total effect value of 0.281. The indirect effect of Perceived ease of use on Actual use behavior value is 0.271. Perceived usefulness has a positive effect on Attitude (0.886). Perceived usefulness has direct (0.401) and indirect (0.426) effects on Actual use behavior, with a total effect of 0.827. Attitude has a positive direct effect on Actual use behavior, and its effect value is 0.481 (Table 5).

This study also examines whether Perceived price is a moderating variable in which Attitudes towards Actual use behavior. The respondents were divided into two groups according to the average score of their perceived price. The path coefficient of the high perceived price group in the model is 0.403, which is less than 0.533 of the low perceived price group. This result supports Hypothesis 10 of this study, which indicates that when users are less sensitive to perceived price, their attitude towards paid video streaming platforms will affect their usage behavior. On the contrary, if users are more sensitive to perceived price, their willingness to subscribe is more likely to be affected by the price. When we examine age by the same means, it is found that there is no difference between high- and low-age groups. This result does not support Hypothesis 11 of this study, and age turns out to have no moderating effect on the relationship between Attitude and Actual use behavior.

Table 5 Effect analysis

Latent independent variable	Latent dependent variable	Direct effect	Indirect effect	Total effect
Personal motivation	Perceived ease of use	0.495***	–	0.495***
System quality		0.336*	–	0.336*
Personal motivation	Perceived usefulness	0.508***	0.168***	0.675***
System quality		0.155($P=0.13$)	0.114**	0.269*
Perceived ease of use		0.338***	–	0.338***
Personal motivation	Attitude	–	0.589***	0.589***
System quality		–	0.232*	0.232*
Perceived ease of use		− 0.018($P=0.942$)	0.300***	0.281**
Perceived usefulness		0.886***	–	0.886***
Personal motivation	Actual use behavior	–	0.554***	0.554***
System quality		–	0.219*	0.219*
Perceived ease of use		–	0.271***	0.271***
Perceived usefulness		0.401**	0.426***	0.827**
Attitude		0.481***	–	0.481***

*Represents $p<0.05$, **represents $p<0.01$, ***represents $p<0.001$

5 Conclusion

Based on TAM, this research explores the factors that may affect users' willingness to use paid video streaming platforms, and further observes whether Perceived price will affect usage attitudes and actual use behaviors.

5.1 Constructs interpretations

Consumers expect to kill their time and meet their needs for entertainment by using paid video streaming platforms, while idolatry or keeping up with current events isn't most of the users' major concerns. The results find that Personal motivation will directly affect users' Perceived usefulness and Perceived ease of use of paid video streaming platforms, and indirectly affect usage Attitudes and Actual use behaviors. OTT platforms with higher content diversity are more appealing to consumers to subscribe. However, users still have doubts about the security of paid video streaming platforms, which therefore reduces their willingness to use them. The research results show that System quality directly affects users' Perceived ease of use of paid video streaming platforms, has no significant direct impact on Perceived usefulness, but will indirectly affect Perceived usefulness through Perceived ease of use. System quality also indirectly affects Attitudes and Actual use behavior. The research results point out that Perceived ease of use has a positive effect on Perceived usefulness. Although Perceived ease of use has no significant direct effect on usage Attitude, it indirectly affects Attitude and Actual use behavior through Perceived usefulness. Gender positively affects Attitudes and Actual use behaviors, while Attitudes positively and directly affect Actual use behavior. In terms of the direct effect of perceived constructs on usage attitudes, Perceived usefulness has a greater effect than Perceived ease of use. For the paid video streaming platforms, a useful platform is more effective than a user-friendly platform. If a platform has no practical functions, consumers are less likely to use it even if the platform operation is simple. Users usually find their own way to overcome a useful platform with a complicated interface. Perceived price plays an interfering role in the relationship between attitude and actual use behavior. Whether a user subscribes to a streaming platform depends on the price. The attitude of the video streaming platform will affect its usage behavior. On the contrary, if the user's perception of the perceived price is "expensive", the willingness to subscribe is more likely to be affected by the price. Additionally, some users feel that the fees charged by paid video streaming platforms will make them feel burdened. In this study, age does not contribute to any difference between attitudes and actual use behaviors. A positive attitude toward paid video streaming platforms, regardless of age, has a positive association with continuous usage behavior in the future. However, if the age is complicated into "the elderly" and "the young" two groups, only 30% of those who have never used paid video streaming platforms constitute the younger age group, while more than 60% of them make up the elderly age group. It can be suggested that, although age cannot directly affect usage attitudes and actual usage behaviors, it can still be applied as a rough distinction too judge usage intentions.

5.2 Implications from a practical perspective

In terms of Personal motivation, more consumers expect to kill the time and meet their needs for entertainment through using paid video streaming platforms. Convenience is often seen as the most important factor which affects consumers' willingness to use (Hino 2015). In terms of System quality, users pay the most attention to the convenience of the platform and the diversity of the content of the platform. Therefore, in order to increase the frequency for users to use paid audiovisual streaming platforms, we can refer to the analysis results by three aspects: improving its entertainment effect, increasing the convenience of the platform, and continuously diversifying video content. Next, the results show that when users are less sensitive to perceived price, their attitude towards paid video streaming platforms will affect their use behavior. When the user is more sensitive to the perceived price, the willingness to subscribe is more likely to be affected by the price. According to Chang and Wildt (1994), perceived price is the subjective perception and response of consumers to the price when they are purchasing service or a product. Perceived price will be positively affected by pricing goals and quality perception; and negatively affected by reference price. Therefore, if OTT operators plan to reduce the effect of perceived price on actual use behavior, they can improve users' perception of the quality of the platform rather than lowering the pricing of the platform. The findings that most of the users prefer recurring charges plan, followed by the hybrid plan, are in resonance with the fact that paid video streaming platforms are most frequently used by Taiwanese users.

5.3 Discussion

OTT has revolutionized film and television culture and reading habits through innovative technologies. Although the performance has grown rapidly, with fierce competition and stagnant growth, the general method of collecting and analyzing audience behaviors is not enough to gain insight into consumers' attitude toward using and behavioral intention of use, and further grasp their actual system use behaviors. This study uses TAM to understand that perceived ease of use is influenced by personal motivation and system quality, which in turn affects perceived usefulness. And perceived usefulness is influenced by personal motivation and perceived ease of use, which in turn affects attitudes and actual use behavior. By confirming the model path, the study found that the actual use behavior of OTT users depends on attitude, and attitude is mainly derived from personal motivation and perceived usefulness supported by perceived ease of use. In other words, the greater the usefulness of OTT, the more positive attitude towards use will be. The personal motivation and system quality of this study provide further theoretical reference for general collection and audience behaviors. OTT operators can ponder about how to increase the perceived usefulness of consumers, thereby increasing the enthusiasm of consumers. This research also gives OTT operators a far-reaching direction when developing or providing new content and systems.

5.4 Limitations and future research

Regarding the descriptive statistics of the respondents, it can be suggested that more than 70% of the respondents are female, and the majority of users are 19–25 years old. Since the online questionnaire may also be one of the reasons for the insignificant effect of age interference in this study, it is recommended to conduct research and comparison on users of different ages in the future or design a survey through sampling to infer users of all paid video streaming platforms. As for the model, Davis (1989) points out that intrinsic motivation has not been paid enough attention in this research, and there is a complex relationship between emotional attitudes and user behaviors. It is still necessary to study how to incorporate intrinsic motivation into TAM model. Although the model of this study shows that the degree of accessories is satisfying enough, the validity of this model still needs to be further verified by future research.

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Declarations

Conflict of interest All authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript. The suggested reviewers are totally independent and not connected to this article in any way.

References

Ahn, T., Ryu, S., Han, I.: The impact of the online and offline features on the user acceptance of Internet shopping malls. Electron. Commer. Res. Appl. **3**(4), 405–420 (2004)

Amoroso, S., Pattuglia, S., Khan, I.: Do Millennials share similar perceptions of brand experience? A clusterization based on brand experience and other brand-related constructs: the case of Netflix. J. Market. Anal. **9**(1), 33–43 (2021)

Bagozzi, R.P., Yi, Y.: On the evaluation of structural equation models. J. Acad. Mark. Sci. **16**(1), 74–94 (1988)

BEREC.: Report on OTT services (2016). https://berec.europa.eu/eng/document_register/subject_matter/berec/reports/5751-berec-report-on-ott-services

Bilbil, E.T.: Methodology for the regulation of Over-the-top (OTT) Services: the need of a multi-dimensional perspective. Int. J. Econ. Financ. Issues **8**(1), 101 (2018)

Bollen, K.A.: Structural Equations with Latent Variables, vol. 210. Wiley, New York (1989)

Bollen, K.A., Stine, R.A.: Bootstrapping goodness-of-fit measures in structural equation models. Sociol. Methods Res. **21**(2), 205–229 (1992)

Cha, J.: Usage of video sharing websites: drivers and barriers. Telemat. Inform. **31**(1), 16–26 (2014)

Chang, T.Z., Wildt, A.R.: Price, product information, and purchase intention: an empirical study. J. Acad. Mark. Sci. **22**(1), 16–27 (1994)

Chatterjee, S.: Factors impacting behavioral intention of users to adopt IoT in India: from security and privacy perspective. Int. J. Inf. Secur. Priv. (IJISP) **14**(4), 92–112 (2020)

Chatterjee, S.: Antecedence of attitude towards IoT usage: a proposed unified model for IT Professionals and its validation. Int. J. Hum. Capit. Inf. Technol. Prof. (IJHCITP) **12**(2), 13–34 (2021a)

Chatterjee, S.: Impact on addiction of online platforms on quality of life. Australas. J. Inf. Syst. 25 (2021b)

Chatterjee, S., Bhattacharjee, K.K., Tsai, C.W., Agrawal, A.K.: Impact of peer influence and government support for successful adoption of technology for vocational education: a quantitative study using PLS-SEM technique. Qual. Quant. **55**(6), 2041–2064 (2021a)

Chatterjee, S., Rana, N.P., Dwivedi, Y.K., Baabdullah, A.M.: Understanding AI adoption in manufacturing and production firms using an integrated TAM-TOE model. Technol. Forecast. Soc. Chang. **170**, 120880 (2021b)

Chong, A.Y.L.: Mobile commerce usage activities: the roles of demographic and motivation variables. Technol. Forecast. Soc. Chang. **80**(7), 1350–1359 (2013)

Chung, N., Koo, C., Kim, J.K.: Extrinsic and intrinsic motivation for using a booth recommender system service on exhibition attendees' unplanned visit behavior. Comput. Hum. Behav. **30**, 59–68 (2014)

Dasgupta, D., Grover, D.: Understanding adoption factors of over-the-top video services among millennial consumers. Int. J. Comput. Eng. Technol. **10**(1), 61–71 (2019)

Davis, F. D.: Perceived usefulness, perceived ease of use, and user acceptance of information technology. MIS Q., 319–340 (1989)

Davis, F.D., Bagozzi, R.P., Warshaw, P.R.: User acceptance of computer technology: a comparison of two theoretical models. Manag. Sci. **35**(8), 982–1003 (1989)

Diddi, A., LaRose, R.: Getting hooked on news: uses and gratifications and the formation of news habits among college students in an Internet environment. J. Broadcast. Electron. Media **50**(2), 193–210 (2006)

eMarketer.: US Cord-Cutter Households, 2019–2024 (2020). https://www.emarketer.com/content/pay-tv-suffers-historic-cord-cutting

Hair, J., Black, W., Babin, B., Anderson, R.: Multivariate Data Analysis. Prentice Hall, London (2009)

He, X.: Survey report on idol worship among children and young people. Chin. Educ. Soc. **39**, 84–103 (2006)

Hino, H.: TV today, mobile TV tomorrow? Extrapolating lessons from Israeli consumers' adoption of innovative TV viewing technology. Int. J. Media Manag. **17**(2), 69–92 (2015)

Holzwarth, M., Janiszewski, C., Neumann, M.M.: The influence of avatars on online consumer shopping behavior. J. Mark. **70**(4), 19–36 (2006)

Kashyap, R., Bojanic, D.C.: A structural analysis of value, quality, and price perceptions of business and leisure travelers. J. Travel Res. **39**(1), 45–51 (2000)

Kim, D., Johnson, T.J.: Political blog readers: predictors of motivations for accessing political blogs. Telemat. Inform. **29**(1), 99–109 (2012)

Kim, M.S., Kim, E., Hwang, S., Kim, J., Kim, S.: Willingness to pay for over-the-top services in China and Korea. Telecommun. Policy **41**(3), 197–207 (2017)

Kim, T.Y., Shin, D.H.: The survival strategy of branded content in the over-the-top (OTT) environment: eye-tracking and Q-methodology approach in digital product placement. Telemat. Inform. **34**(7), 1081–1092 (2017)

Koul, S., Ambekar, S.S., Hudnurkar, M.: Determination and ranking of factors that are important in selecting an over-the-top video platform service among millennial consumers. Int. J. Innov. Sci. **13**(1), 53–66 (2020)

Kwon, Y., Park, J., Son, J.Y.: Accurately or accidentally? Recommendation agent and search experience in over-the-top (OTT) services. Internet Res. **31**(2), 562–586 (2020)

Leung, L.: Impacts of Net-generation attributes, seductive properties of the Internet, and gratifications-obtained on Internet use. Telemat. Inform. **20**(2), 107–129 (2003)

Lichtenstein, D.R., Ridgway, N.M., Netemeyer, R.G.: Price perceptions and consumer shopping behavior: a field study. J. Mark. Res. **30**(2), 234–245 (1993)

Liebermann, Y., Stashevsky, S.: Perceived risks as barriers to Internet and e-commerce usage. Qual. Mark. Res. Int. J. **5**(4), 291–300 (2002)
Lombard, M., Ditton, T.: At the heart of it all: the concept of presence. J. Comput.-Mediated Commun. **3**(2), JCMC321 (1997)
Lotz, A.: Original or exclusive? Shifts in television financing and distribution shift meanings. Antenna: Responses to Media and Culture, 1 January. Accessed 3 May 2017 http://blog.commarts.wisc.edu/2016/01/01/original-or-exclusive-shifts-in-television-financing-and-distributionshift-meanings/ (2016)
Maltby, J., Day, L., McCutcheon, L.E., Gillett, R., Houran, J., Ashe, D.D.: Personality and coping: a context for examining celebrity worship and mental health. Br. J. Psychol. **95**, 411–428 (2004)
McCutcheon, L.E., Ashe, D.D., Houran, J., Maltby, J.: A cognitive profile of individuals who tend to worship celebrities. J. Psychol. **137**, 309–322 (2003)
McQuail, D.: Media Performance: Mass Communication and the Public Interest, vol. 144. Sage, London (1992)
MPAA.: 2018 THEME REPORT (2018). https://www.motionpictures.org/research-docs/2018-theatrical-home-entertainment-market-environment-theme-report/
Mulla, T.: Assessing the factors influencing the adoption of over-the-top streaming platforms: a literature review from 2007 to 2021. Telemat. Inform. **69**(C), 101797 (2022)
Papacharissi, Z., Rubin, A.M.: Predictors of internet use. J. Broadcast. Electron. Media **44**(2), 175–196 (2000)
Park, E.A.: Business strategies of Korean TV players in the age of over-the-top (OTT) video service. Int. J. Commun. **12**, 22 (2018)
Park, J., Song, Y., Teng, C.I.: Exploring the links between personality traits and motivations to play online games. Cyberpsychol. Behav. Soc. Netw. **14**(12), 747–751 (2011)
Pornsakulvanich, V., Dumrongsiri, N.: Internal and external influences on social networking site usage in Thailand. Comput. Hum. Behav. **29**(6), 2788–2795 (2013)
Redmond, W.H.: The potential impact of artificial shopping agents in e-commerce markets. J. Interact. Mark. **16**(1), 56–66 (2002)
Sadana, M., Sharma, D.: How over-the-top (OTT) platforms engage young consumers over traditional pay television service? An analysis of changing consumer preferences and gamification. Young Consum. **22**(3), 348–367 (2021)
Samuelson, P.: Digital media and the changing face of intellectual property law. Rutgers Comput. Tech. LJ **16**, 323 (1990)
Sharma, K., Lulandala, E.E.: OTT platforms resilience to COVID-19–a study of business strategies and consumer media consumption in India. Int. J. Org. Anal. (2022). https://doi.org/10.1108/IJOA-06-2021-2816
Shin, J., Park, Y., Lee, D.: Strategic management of over-the-top services: focusing on Korean consumer adoption behavior. Technol. Forecast. Soc. Chang. **112**, 329–337 (2016)
Singh, P.: New media as a change agent of Indian television and cinema: a study of over the top platforms. J. Content Commun. Commun. **9**(1), 131–137 (2019)
Snyman, J.H., Gilliard, D.J.: The streaming television industry: mature or still growing? J. Mark. Dev. Compet. **13**(4), 94–105 (2019)
Sujata, J., Sohag, S., Tanu, D., Chintan, D., Shubham, P., Sumit, G.: Impact of over the top (OTT) services on telecom service providers. Indian J. Sci. Technol. **8**(S4), 145–160 (2015)
Swanson, S.R., Davis, J.C., Zhao, Y.: Art for art's sake? An examination of motives for arts performance attendance. Nonprofit Volunt. Sect. Q. **37**(2), 300–323 (2008)
Taipei Association of Advertising Agencies. 2020 Media Book, http://www.taaa.org.tw/
Taiwan Network Information Center (TWNIC).: 2019 Taiwan Internet Report (2019). https://report.twnic.tw/2019/
Taylor, S., Todd, P.A.: Understanding information technology usage: a test of competing models. Inf. Syst. Res. **6**(2), 144–176 (1995)
Torkzadeh, G., Koufteros, X., Pflughoeft, K.: Confirmatory analysis of computer self-efficacy. Struct. Equ. Model. **10**(2), 263–275 (2003)
Van der Wurff, R.: Business magazine market performance: magazines for the agricultural, business services, and transportation sectors in the Netherlands. J. Media Econ. **18**(2), 143–159 (2005)
Vlachos, P.A., Tsamakos, A., Vrechopoulos, A.P., Avramidis, P.K.: Corporate social responsibility: attributions, loyalty, and the mediating role of trust. J. Acad. Mark. Sci. **37**(2), 170–180 (2009)
Wayne, M.L.: Netflix, Amazon, and branded television content in subscription video on-demand portals. Media Cult. Soc. **40**(5), 725–741 (2018)
Wixom, B.H., Todd, P.A.: A theoretical integration of user satisfaction and technology acceptance. Inf. Syst. Res. **16**(1), 85–102 (2005)

Wolfinbarger, M., Gilly, M.C.: Shopping online for freedom, control, and fun. Calif. Manag. Rev. **43**(2), 34–55 (2001)

Wolk, A.: Over the top: how the internet is (slowly but surely) changing the television industry. CreateSpace Independent Publishing Platform, North Charleston, SC (2015)

Yoo, J., Lee, J., Lee, D.: A verification of motivations for over-the-top binge and short viewing of audio-visual content. New Rev. Hypermedia Multimed. **26**(3–4), 93–122 (2020)

Zeithaml, V.A.: Consumer perceptions of price, quality, and value: a means-end model and synthesis of evidence. J. Mark. **52**(3), 2–22 (1988)